Log-radius response versus directional ellipticity, split by galaxy properties

True R_e (arcsec)

Property interval
0.5–0.593
0.593–0.728
0.728–0.952
0.952–1.5

True magnitude

Property interval
18–23.4
23.4–24.3
24.3–25
25–25.8

Blending response R_{blend}

Property interval
-2.09–-0.00531
-0.00531–0.00965
0.00965–0.0882
0.0882–3.66

Measured zero-leg radius (arcsec)

Property interval
0.242–0.833
0.833–0.95
0.95–1.11
1.11–30.1

Measured zero-leg magnitude

Property interval
17.7–23.2
23.2–24.1
24.1–24.8
24.8–28

Sersic index n

Property interval
0.513–0.847
0.847–1.17
1.17–1.75
1.75–5.85

$\Delta \ln r / \Delta g$ (per unit shear)

Intrinsic $e_\parallel$ (along applied shear)

1,120,160 matched galaxies; 40 validation cases; true Re > 0.5 arcsec; both legs detected/usable; no measured selection cuts.
Curves: mean response in 12 common ellipticity bins; bars: 1 SE from 2,000 whole-case bootstrap resamples.
Colors split each property into quartiles (ties merged). Other properties are not held fixed; all finite responses, including tails, enter means.